

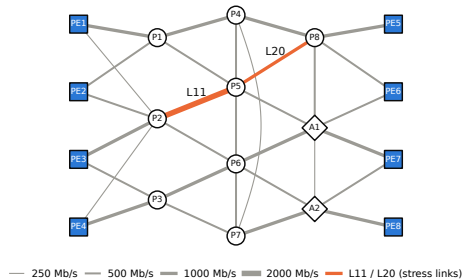
Persistent state is not enough

A controlled study of planning horizons in incremental MPLS traffic engineering

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1. Problem: incremental MPLS traffic engineering

- ▶ 18 routers, 64 directed links, 17 demands with SLAs, 4 candidate LSPs each
- ▶ every 5 min: move *one* demand to another candidate, or do nothing
- ▶ moves cost reconfiguration effort; 3-interval hold-down after a move
- ▶ action mask: failed links, hold-down, protected-class capacity
- ▶ reward: delivery, disconnection, SLA severity, utilization, overload, minus move costs



2. Why RL — and why sequentiality should not be assumed

- ▶ routes **persist**, moves lock a demand, protected-class legality couples demands $\Rightarrow$ formally an MDP
- ▶ but a move carries traffic *in the interval it is made*, traffic is exogenous
- ▶ recent TE systems treat TE as one-shot (Teal, DOTE) — in a setting *without* persistent state
- ▶ **Question:** here, does a learner that plans ($\gamma = 0.995$) beat one that only predicts the immediate reward ($\gamma = 0$)? Why?

3. The surprising observation, reproduced

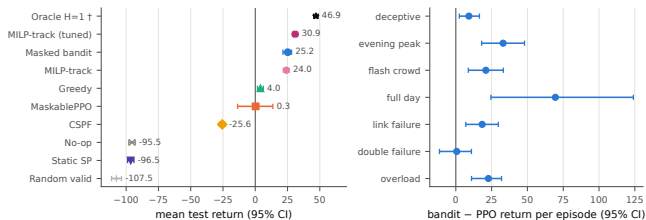
- ▶ closed predecessor study: bandit 18.2 vs PPO 9.0 (holdout return, 3 roots)
- ▶ retrained everything on different hardware, byte-identical training traffic:
 - ▶ baselines: **exact** ($3 \cdot 10^{-14}$)
 - ▶ bandit: **reproduces** (18.0)
 - ▶ PPO: **does not** (-12.0); per-root shifts up to 41 points
 - ▶ “PPO wins the deceptive scenario”: does not reproduce

Learner	Root	Selected (h./r.)	Hist.	Repr.
Bandit	42	250k / 400k	16.1	14.7
Bandit	271828	400k / 350k	18.0	22.2
Bandit	314159	300k / 250k	20.5	17.2
PPO	42	250k / 350k	6.6	-20.1
PPO	271828	150k / 400k	12.2	-29.0
PPO	314159	350k / 200k	8.3	13.0

4. Methodology

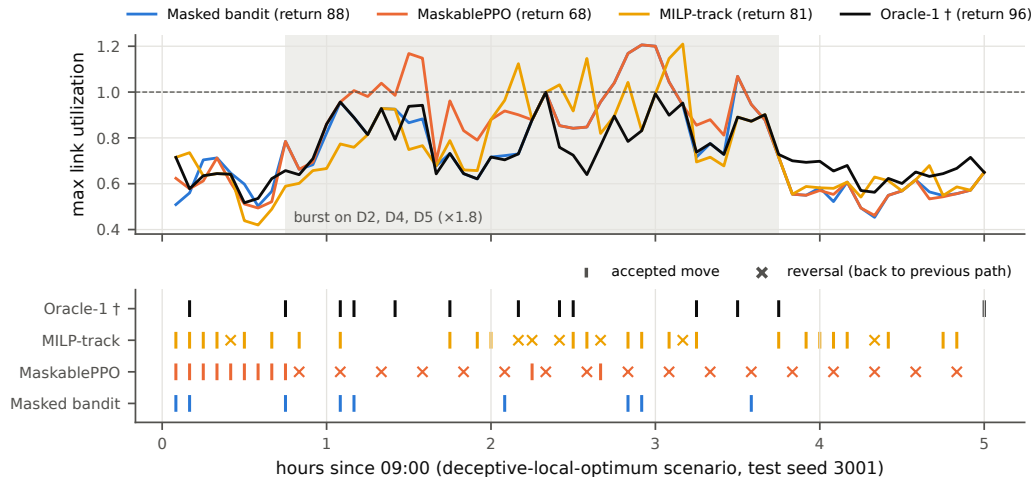
- ▶ same observation (604), mask, budget (400k transitions), paired traffic
- ▶ select on validation seeds, report on 140 disjoint test episodes; 5 training roots
- ▶ two-stage bootstrap CIs over roots and episodes
- ▶ references: static, greedy, CSPF, **MILP-track** (per-interval min-max-util), clairvoyant Oracle- H
- ▶ controls: PPO $\gamma \in \{0, 0.9, 0.995\}$; masked Q-learner $\gamma \in \{0, 0.5, 0.9, 0.99\}$ ($\gamma = 0$ = bandit, bit-identical); PPO tuning; delayed-effect variant; time-of-day observation

5. Result: bandit $>$ PPO; a tuned per-interval optimizer beats both



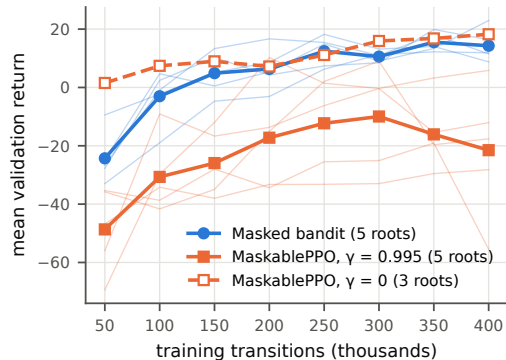
- ▶ bandit - PPO = **24.9**,
 t -interval over roots [3.4, 46.4],
positive on 5/5 roots
- ▶ bandit - MILP-track (untuned)
= 1.2 [-4.0, 6.3]
- ▶ MILP-track tuned on
validation: **30.9**, ahead of the
bandit on every root
- ▶ reroutes/h: bandit 2.30, MILP
5.83, PPO 6.38

6. Mechanism: PPO flaps at hold-down expiry



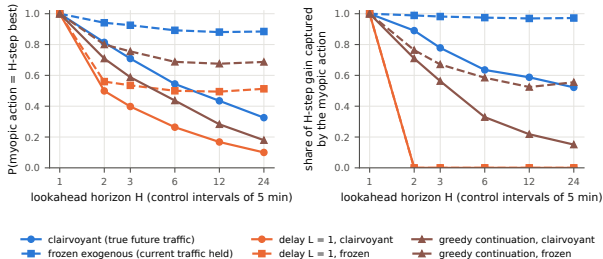
Utility: bandit 28.0, PPO 18.4; move costs: bandit -2.8, PPO -18.1

7. Attribution: the horizon, not the algorithm (so far)



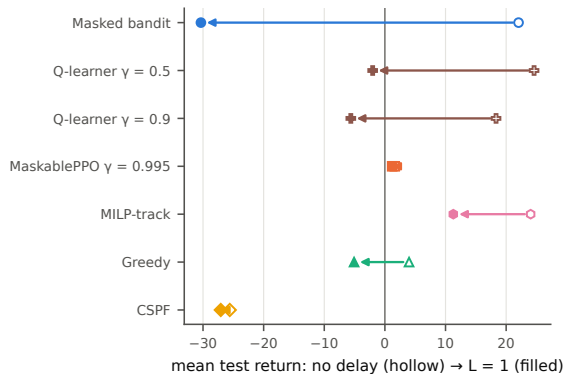
- ▶ PPO with $\gamma = 0$: **23.6** vs bandit 24.2 on the same roots
- ▶ difference -0.6 $[-5.2, 2.7]$
- ▶ flapping disappears
- ▶ return falls with γ in both families (PPO $\gamma = 0.9$: 15.4; Q $\gamma = 0.99$: 3.3)
- ▶ tuned PPO (best of 8 on one root) below the bandit on all 4 fresh roots
- ▶ hypothesis: small one-off costs drowned by long-horizon advantage noise

8. How sequential is the problem?



- ▶ re-planning with perfect foresight: 3-step lookahead adds only 2.9
- ▶ open loop, true future: myopic = 24-step best in 33 % of states
- ▶ traffic frozen: **88 %**
- ▶ open-loop non-myopic value $\approx$ anticipating exogenous change, not provided by the observation
- ▶ **but:** with a greedy controller reacting after the first move, agreement falls to 18 % (frozen 69 %) — first moves shape later value; the learners did not exploit it

9. When does the myopic learner fail? Delay the effect of a move



- ▶ move requested at t , active from $t + 1$: the immediate reward contains only its cost
- ▶ bandit: 22.0 $\rightarrow$ -30.3
- ▶ PPO: 1.6 $\rightarrow$ 1.2 (mean; per-root shifts in both directions)
- ▶ PPO — bandit under delay: 31.5 (mean), ahead on 2/2 roots
- ▶ MILP-track: 24.0 $\rightarrow$ 11.3 — still ahead of every learner
- ▶ shows: decision-time regression fails; not that long-horizon RL is needed
- ▶ the rollout diagnostic flags it without training: myopic choice captures 0% of the 24-step gain (97% at $L = 0$)

10. Clairvoyance: how much is lookahead worth?

- ▶ Oracle- H : enumerates every legal move with the true future, no-op continuation
- ▶ Oracle-1 48.6, Oracle-3 51.4, Oracle-6 50.0 (5 test seeds)
- ▶ Oracle-3 $-$ Oracle-1 = 2.9 [1.5, 4.2]; Oracle-6 $-$ Oracle-1 = 1.4 [0.5, 2.5]
- ▶ bandit $-$ Oracle-1 = -21.7 : **knowing the next interval** is worth far more than planning beyond it

11. Limitations

- ▶ flow-level model, exogenous inelastic traffic, instantaneous control plane
- ▶ one topology, scripted scenarios, one reward; 5 (main) / 2–3 (ablation) roots
- ▶ single-move-per-interval formulation is a design choice

12. Take-aways and future work

- ▶ measure the value of lookahead before choosing a sequential learner: the rollout diagnostic separated the base and the delayed regime without training
- ▶ learner noise within a root is large (10.4 points): small effects need many roots
- ▶ a tuned per-interval optimizer is the baseline to beat; a myopic learner is next
- ▶ long horizons can hurt by hiding small immediate costs
- ▶ future: delayed actuation and elastic traffic; packet-level validation; GNN policies; larger topologies; real traffic matrices